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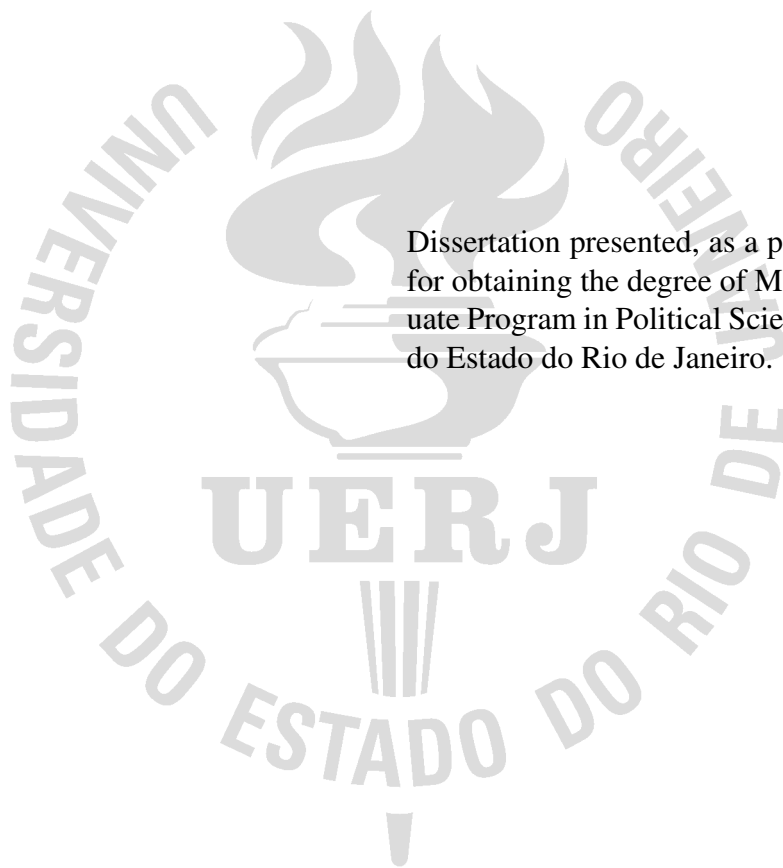
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ABSTRACT

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RESUMO

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INTRODUCTION

To the worm that first gnawed the cold flesh of my corpse I dedicate these Posthumous Memoirs as a fond remembrance (Assis, 1881).

In statistics, “all models are wrong, but some are useful” (Box, 1976). A model is a deliberate simplification of reality, and the craft of modelling lies in choosing which aspects to keep. Gelman; Hill (2007) show how regression and hierarchical models negotiate this tension between parsimony and realism.

The quality of our conclusions depends on both the model and the data. A striking example is the forecasting of U.S. elections from a sample of Xbox users — heavily biased relative to the electorate — which, after suitable statistical adjustment, produced estimates competitive with traditional polls (Wang *et al.*, 2015).

1 ORDINARY LEAST SQUARES

Consider the linear regression model in matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1)$$

where $\mathbf{y}$ is the $n \times 1$ response vector, $\mathbf{X}$ is the $n \times p$ design matrix, $\boldsymbol{\beta}$ is the coefficient vector and $\boldsymbol{\varepsilon}$ is the error vector.

1.1 Closed-form derivation

The ordinary least squares estimator minimises the residual sum of squares:

$$S(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Expanding the product and using that $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}$ is a scalar (hence equal to its transpose),

$$S(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

The first-order condition follows by differentiating with respect to $\boldsymbol{\beta}$ and setting the gradient to zero:

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} \stackrel{!}{=} \mathbf{0}.$$

Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, we solve for the estimator in its celebrated closed form:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (2)$$

1.2 Illustration

Figure 1 shows a set of points (x_i, y_i) and the line fitted by Equation 2. The estimated intercept and slope are $\hat{\beta}_0 \approx 1.78$ and $\hat{\beta}_1 \approx 0.79$, respectively.

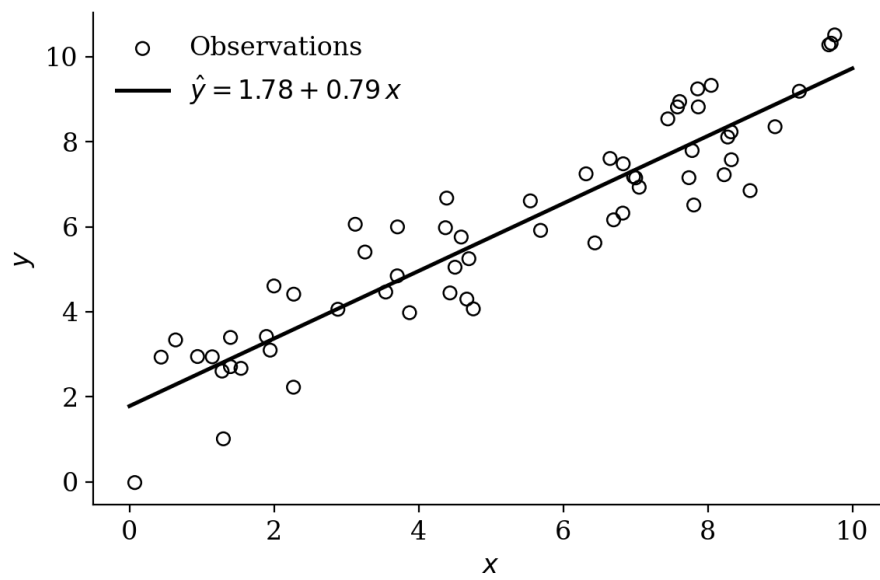


Figure 1 - Regression line fitted by ordinary least squares. Source: prepared by the author (2026).

CONCLUSION

Models are useful when they help us learn from data (Box, 1976), and the closed form of OLS in Equation 2 is the starting point for richer extensions, such as the hierarchical models of Gelman; Hill (2007).

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